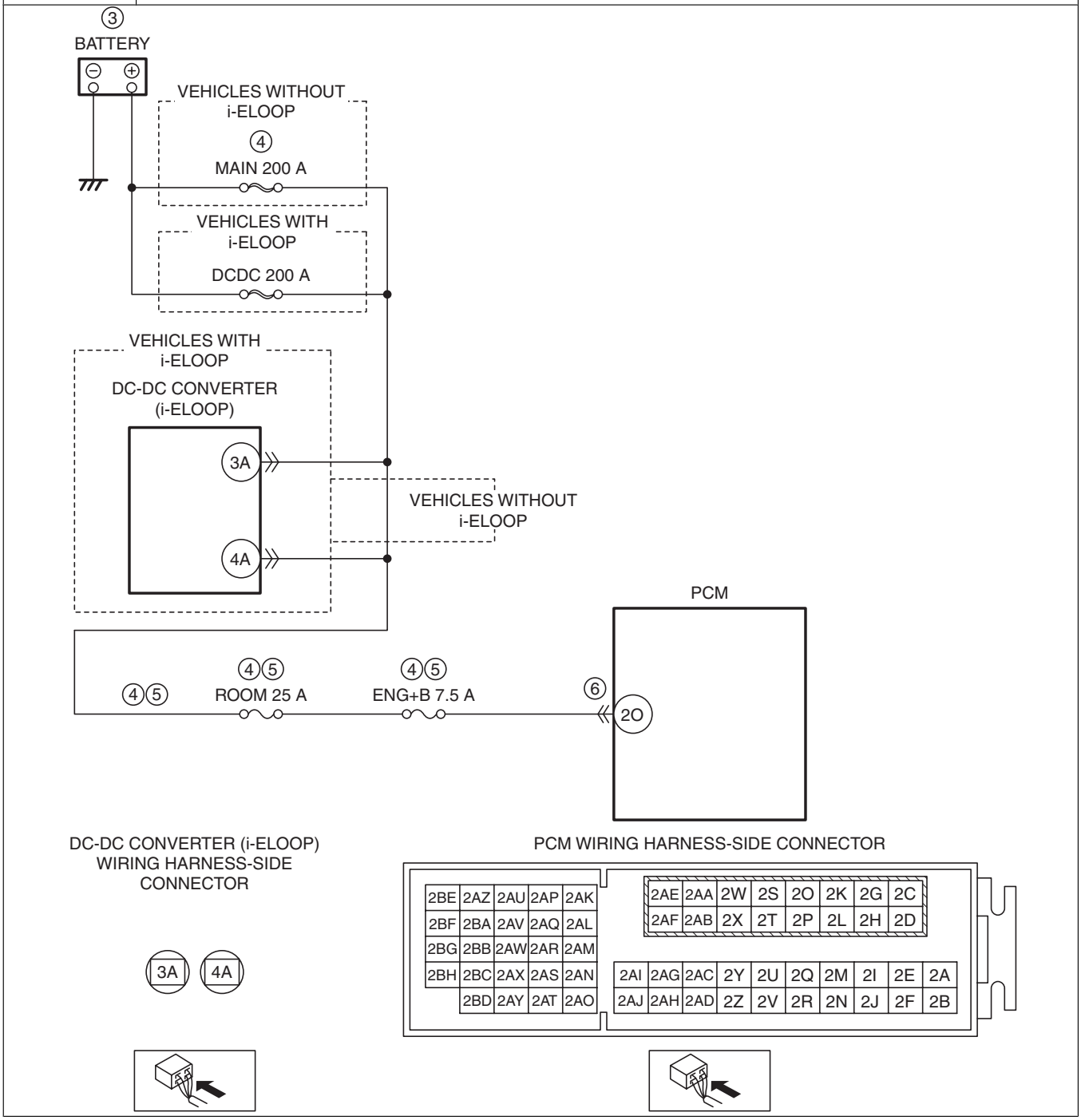
DTC P2507:00 [SKYACTIV-D 1.5]

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DTC P2507:00	PCM battery voltage low input
DETECTION CONDITION	• When the following conditions are met, a condition in which the backup power voltage is **less than 50%** of the battery voltage continues for **10 s**. **MONITORING CONDITIONS** — Battery voltage: **10—16 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Battery malfunction • Short to ground or open circuit in backup voltage circuit (without i-ELOOP) — Short to ground in wiring harness between MAIN 200 A fuse and PCM terminal 2O — MAIN 200 A fuse and/or ROOM 25 A fuse and/or ENG+B 7.5 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and PCM terminal 2O • Short to ground or open circuit in backup voltage circuit (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and PCM terminal 2O — ROOM 25 A fuse and/or ENG+B 7.5 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and PCM terminal 2O • PCM connector or terminals malfunction • PCM malfunction

DTC P2507:00	PCM battery voltage low input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.

STEP	INSPECTION		ACTION
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT BATTERY** • Switch the ignition off. • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Recharge or replace the battery, then go to Step 7. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Vehicles with i-ELOOP: • Go to Step 5. Vehicles without i-ELOOP: • Go to the next step.
4	**INSPECT BACKUP VOLTAGE CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Reconnect all disconnected connectors. • Access the VPWR PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Verify the VPWR PID value. • Is the VPWR PID value **B+**?	Yes	Go to Step 6.
		No	Inspect the MAIN 200 A fuse, ROOM 25 A fuse and ENG+B 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.

STEP	INSPECTION		ACTION
5	**INSPECT BACKUP VOLTAGE CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Reconnect all disconnected connectors. • Access the VPWR PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Verify the VPWR PID value. • Is the VPWR PID value **B+**?	Yes	Go to the next step.
		No	Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the ROOM 25 A fuse and ENG+B 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ROOM 25 A fuse and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Go to Step 7.
6	**INSPECT PCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.